

Homogeneous Embeddings of Cycles in Graphs

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Abstract

If G and H are vertex-transitive graphs, then the framing number $fr(G, H)$ of G and H is defined as the minimum order of a graph every vertex of which belongs to an induced G and an induced H . This paper investigates $fr(C_m, C_n)$ for $m < n$. We show first that $fr(C_m, C_n) \geq n + 2$ and determine when equality occurs. Thereafter we establish general lower and upper bounds which show that $fr(C_m, C_n)$ is approximately the minimum of $n - m + 2\sqrt{n}$ and $n + n/m$.

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1 Introduction

Chartrand, Gavlas, and Schultz [1] introduced the framing number of a graph. A graph G can be *homogeneously embedded* in a graph H if for every vertex x of G and every vertex y of H , there exists an embedding of G in H as an induced subgraph with x at y . A graph F of minimum order in which G can be homogeneously embedded is called a *frame* of G , and the order of F is called the *framing number* $fr(G)$ of G . In [1] it is shown that a frame exists for every graph, although a frame need not be unique.

Theorem A (Chartrand et al. [1]) *Every graph has a frame.*

Theorem B (Chartrand et al. [1]) *If a graph G can be homogeneously embedded in a graph H , then*

$$\Delta(G) \leq \delta(H) \leq \Delta(H) \leq |V(H)| - |V(G)| + \delta(G).$$

Chartrand, Gavlas, and Schultz [1] extended the concept of framing numbers to more than one graph. For graphs G_1 and G_2 , the framing number $fr(G_1, G_2)$ of G_1 and G_2 is defined as the minimum order of a graph F such that G_i ($i = 1, 2$) can be homogeneously embedded in F . The graph F is called a frame of G_1 and G_2 . Then $fr(G_1, G_2)$ exists and, in fact, $fr(G_1, G_2) \leq fr(G_1 \cup G_2)$.

Theorem C (Chartrand et al. [1]) *For any collection $G_1, G_2, \dots, G_m$ of graphs, there exists an integer N such that for each integer $n \geq N$, there is a graph H of order n in which each G_i ($1 \leq i \leq m$) can be homogeneously embedded, while for each integer $n < N$, no such graph exists.*

Results involving frames and framing numbers of graphs have been presented by, among others, Chartrand, Gavlas, and Schultz [1], Chartrand, Henning, Hevia, and Jarrett [2], Entringer, Goddard, and Henning [4], Gavlas, Henning, and Schultz [5], Goddard, Henning, Oellermann, and Swart [6, 7], and Henning [8].

A special case of this concept is when the graphs are vertex-transitive. If G is vertex-transitive, then G can be homogeneously embedded in H if every vertex of H lies in an induced subgraph of H which is a copy of G . The framing number of a single vertex-transitive graph is trivially its order. In [4] one result of this kind is presented: viz. the framing number of a complete graph and an independent set.

Theorem D (Entringer et al. [4]) *For integers $m, n \geq 2$,*

$$fr(K_m, \bar{K}_n) = n + m - 2 + \left\lceil 2\sqrt{(m-1)(n-1)} \right\rceil.$$

In this paper we investigate the framing number of two cycles. In Section 2 we extend the results of Chartrand et al. [1]. For integers $n > m \geq 3$, we show that $fr(C_m, C_n) \geq n + 2$

and we determine those pairs of cycles C_m and C_n which have framing number $n + 2$. In Section 3 we provide constructions which establish upper bounds on $fr(C_m, C_n)$ and in Section 4 we provide arguments which establish lower bounds. As a result it follows that $fr(C_m, C_n)$ is approximately $n + n/m$ for $m \geq \sqrt{6n}$ and approximately $n - m + 2\sqrt{n}$ for $m \leq \sqrt{n}/4$. We discuss this in Section 5.

2 A Simple Lower Bound

Chartrand et al. [1] investigated the framing number $fr(G_1, G_2)$ for several pairs G_1, G_2 of cycles. For small m and n , they established the values of $fr(C_m, C_n)$. The following table summarizes their results.

m	3	3	3	4	4	5
n	4	5	6	5	6	6
$fr(C_m, C_n)$	6	7	8	7	8	8

In this section, we extend these results. We begin with the following lemma.

Lemma 1 *For integers $n > m \geq 3$, $fr(C_m, C_n) \geq n + 2$.*

Proof. By Theorem C, it suffices to show that there is no graph of order $n + 1$ in which C_n and C_m can be homogeneously embedded. Assume, to the contrary, that such a graph H exists. Let $C' : a_1, a_2, \dots, a_n, a_1$ be an induced n -cycle in H , and let x be the vertex of H not in C' . Let C_x be an induced n -cycle containing x . Say C_x is given by $x, a_2, a_3, \dots, a_n, x$. Hence $\deg a_2 = \deg a_n = 3$ and $\deg a_i = 2$ for $i = 3, \dots, n - 1$. However, there is no induced m -cycle containing the vertex a_3 . This produces a contradiction. $\square$

In this section we prove:

Theorem 1 *For integers $n > m \geq 3$, $fr(C_m, C_n) = n + 2$ if and only if*

- (a) $n = m + 1$;
- (b) $n = 2m - 4$, $2m - 3$ or $2m - 2$; or
- (c) $m = 3$ and $n = 5$ or 6 .

By the above lemma, we know that $n + 2$ is a lower bound for the framing number. So we must show that if (m, n) is on the list then there is a graph of order $n + 2$ in which C_m and C_n can be homogeneously embedded, and if (m, n) is not on the list then there is no such graph. We start with the latter.

2.1 Necessary conditions

We may assume that $n \geq 7$, since for smaller values there is nothing to prove. Assume that H has order $n + 2$ and both C_m and C_n can be homogeneously embedded in H . Let C be an induced n -cycle. For notational purposes, we assume that C is oriented. So for any vertex a , a^- denotes the vertex before a and a^+ the vertex after a .

There are two vertices outside C : call these x and y . Let D be an induced n -cycle containing x . There are three possibilities.

- (1) *x and y lie together on D and are adjacent.* Then D consists of a segment S of $n - 2$ vertices of C , together with x and y . Since $n > 4$ there is an internal vertex on S . But any induced cycle containing that vertex has length at least n , since it contains all of S and the ends of S do not have a common neighbour. A contradiction.
- (2) *x and y do not lie together on any n -cycle.* The two vertices outside D are y and one vertex a on C . This means that x is adjacent to a^- and a^+ . Similarly, by considering the induced n -cycle containing y , it follows that y is adjacent to b^- and b^+ for some vertex b on C . The only other possible edges outside C are xa , yb and xy .

If $m \leq 4$ then consider a vertex u on C that is not in the set $\{a^-, a, a^+, b^-, b, b^+\}$. The vertex u cannot lie in a 3- or a 4-cycle. So we may assume that $m \geq 5$.

Now, neither $H - x$ nor $H - y$ contains an m -cycle. So every m -cycle uses both x and y . Consider an induced m -cycle E containing a . Since both a^- and a^+ are neighbours of both x and a , they cannot both be in E . It follows that a must have another neighbour; that neighbour must be y . This means that b is a^- , a or a^+ .

In each case there is a segment S of at least $n - 2$ vertices of C such that the internal vertices of S have degree two. As above, any cycle containing an internal vertex of S has at least n vertices. A contradiction.

- (3) *x and y lie together on D but are not adjacent.* Then the two vertices u and v outside D lie on C , and their removal from C breaks C into two segments; call these S_1 and S_2 . Vertex x is adjacent to exactly one end of S_1 and one end of S_2 , and vertex y is adjacent to the other end of each segment. The only other possible edges are between $\{x, y\}$ and $\{u, v\}$.

Since the two segments S_1 and S_2 combined have $n - 2$ vertices, an m -cycle can contain an interior vertex from at most one of the two segments. Since $n > 6$, at least one segment has an interior vertex. There are two cases:

- (a) There are interior vertices on only one of the segments, say S_1 . Then S_1 contains at least $n - 4$ vertices. Consider an m -cycle E containing one of the vertices of S_2 . If E does not contain S_1 , then it has at most 6 vertices. But any cycle containing S_1 has at least $n - 2$ vertices. So $n - 2 \leq m \leq 6$. Thus we are in Case (a) or (b) of the theorem.

If E does contain S_1 , then there are at least two other vertices in this cycle apart from S_1 , and so it has length $n - 1$. Thus we are in Case (a) of the theorem.

- (b) There are interior vertices on both segments. Then consider an induced m -cycle E that contains S_1 . The cycle E contains at least two vertices outside S_1 , since the ends of S_1 do not have a common neighbour.

Suppose that E contains at least three vertices outside S_1 . We may assume that x is adjacent to the same end of S_1 that u is. Then E cannot contain both u and x ; nor can it contain both v and y (these are both adjacent to the other end of S_1). It follows that E must contain an end of S_2 . Without loss of generality that end is adjacent to u . Then E must contain u , and y , and hence exactly three vertices outside S_1 .

So we have shown that an induced m -cycle that contains S_i has either 2 or 3 more vertices than S_i . But both segments are in induced m -cycles. It follows that if n is even, both S_1 and S_2 have cardinality $n/2 - 1$, and that m is either $n/2 + 1$ or $n/2 + 2$. If n is odd, it follows that one segment has cardinality $(n - 3)/2$ and the other $(n - 1)/2$, and that m is $(n + 3)/2$. Thus we are in Case (b) of the theorem. $\square$

2.2 Frames

The frames for $(m, n) = (3, 5)$ and $(3, 6)$ are given in [1]. For the other cases, examples are presented in Figure 1. With some more work, one can actually determine the nonisomorphic frames in each case. Surprisingly perhaps, the frame is never unique. The entire list can be found in the third author's thesis [9].

3 Upper Bounds

In this section we investigate upper bounds on $fr(C_m, C_n)$.

Theorem 2 *For integers $n > m \geq 3$,*

$$fr(C_m, C_n) \leq \begin{cases} \lceil \frac{4n}{3} \rceil & \text{if } m = 3 \text{ or } 4, \\ n + \frac{n}{m-1} & \text{if } m-1 \mid n, \\ n + \lceil \frac{n}{m-1} \rceil & \text{if } m-1 \mid n-1, \\ n + \lceil \frac{n}{m-1} \rceil + 1 & \text{otherwise.} \end{cases}$$

Proof. $m = 3$ or 4 . Let $k = \lceil n/3 \rceil$. Let G be the graph obtained from the induced n -cycle $C : v_0, v_1, v_2, \dots, v_{n-1}, v_0$ by adding k new vertices $w_0, w_1, \dots, w_{k-1}$ and, for $i = 0, 1, \dots, k-1$, joining w_i to the three vertices v_{3i} , v_{3i+1} and v_{3i+2} (where addition is taken modulo n). Then each vertex of G clearly belongs to a C_3 . Furthermore, the cycle obtained from C by replacing the vertex v_{3i+1} with the vertex w_i (and the edges $w_i v_{3i}$ and $w_i v_{3i+2}$) is an induced n -cycle containing w_i . Hence C_3 and C_n can be homogeneously embedded in the graph G of order $n + k = n + \lceil n/3 \rceil$. Thus $fr(C_3, C_n) \leq \lceil 4n/3 \rceil$.

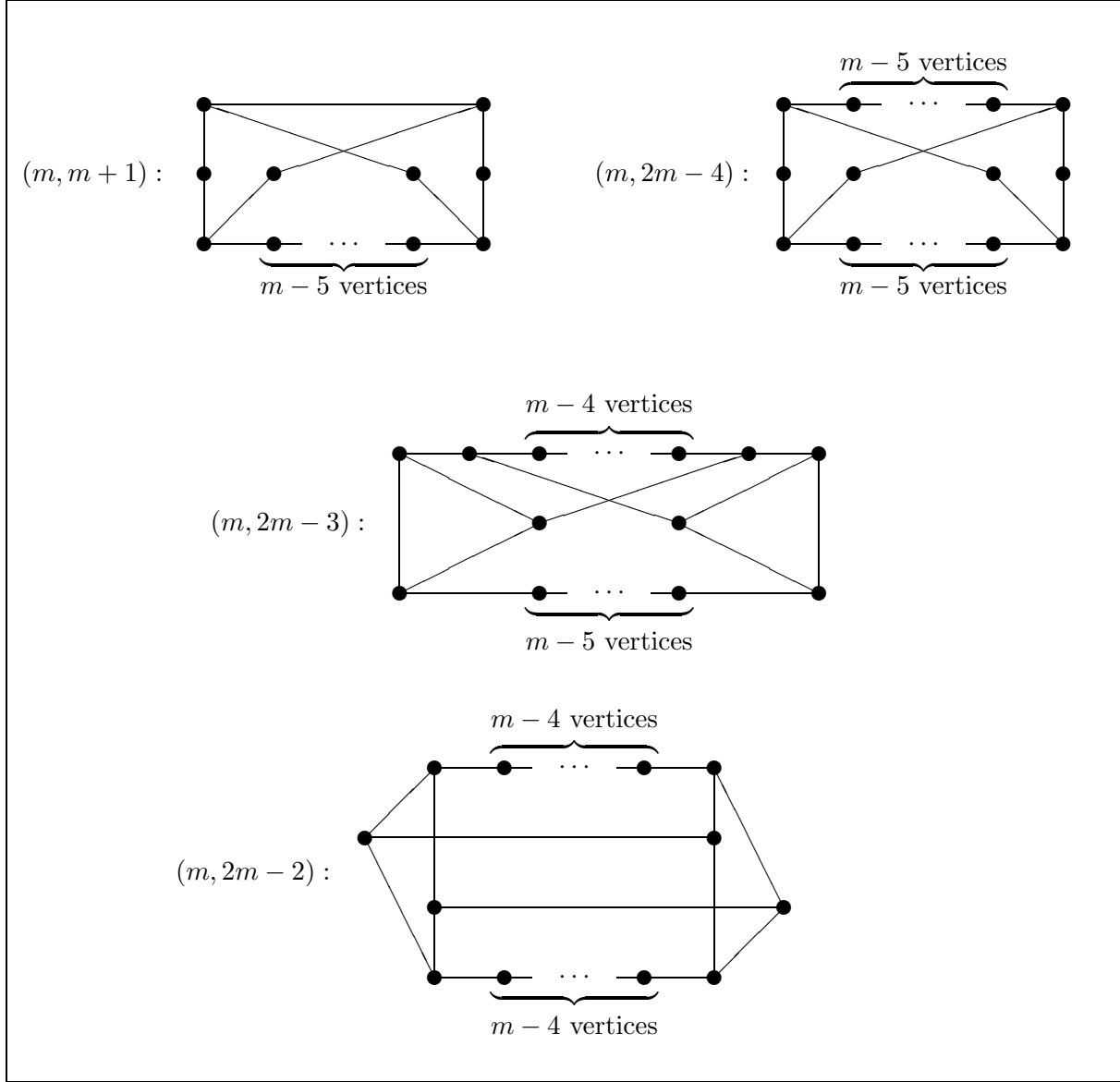


Figure 1: Frames with $n+2$ vertices

Figure 2: The graph $G_{7,18}$

If $m = 4$, then let G' be the graph obtained from G by deleting the edges $w_i v_{3i+1}$ for $i = 0, 1, \dots, k-1$. Then C_4 and C_n can be homogeneously embedded in the graph G' of order $n + k = n + \lceil n/3 \rceil$. Thus $fr(C_4, C_n) \leq \lceil 4n/3 \rceil$.

$m \geq 5$. Let $\ell = \lceil n/(m-1) \rceil$. Let $G_{m,n}$ be the graph obtained from the induced n -cycle $C : v_0, v_1, v_2, \dots, v_{n-1}, v_0$ by adding ℓ new vertices $w_0, w_1, \dots, w_{\ell-1}$ and, for $i = 0, 1, \dots, \ell-1$, joining w_i to the three vertices $v_{i(m-1)-1}$, $v_{i(m-1)+1}$ and $v_{(i+1)(m-1)}$ (where addition is taken modulo n). The graph $G_{7,18}$ is depicted in Figure 2.

Case 1. $m-1 \mid n$.

Thus $n = \ell(m-1)$. Then C_m and C_n can be homogeneously embedded in the graph $G_{m,n}$ of order $n + \ell$. To see this, observe that for $i = 0, 1, \dots, \ell-1$, each vertex w_i belongs to an induced m -cycle, namely $w_i, v_{i(m-1)+1}, v_{i(m-1)+2}, \dots, v_{(i+1)(m-1)}, w_i$. Also, each vertex of C belongs to one of these m -cycles. Furthermore, replacing the vertex $v_{i(m-1)}$ on C with the vertex w_i for all $i = 0, 1, \dots, \ell-1$ produces one induced n -cycle containing every w_i . Consequently, $G_{m,n}$ homogeneously embeds C_m and C_n . Thus, $fr(C_m, C_n) \leq n + n/(m-1)$.

Case 2. $m-1 \mid n-1$.

Thus $n = (\ell-1)(m-1) + 1$. Let $H_{m,n}$ be the graph obtained from $G_{m,n}$ by adding the three edges $w_{\ell-1}w_0$, $w_{\ell-1}v_{(\ell-2)(m-1)}$ and w_1v_0 , and deleting the edge $w_{\ell-1}v_{m-2}$. Then C_m and C_n can be homogeneously embedded in the graph $H_{m,n}$ of order $n + \ell + 1$. (Once again, replacing the vertex $v_{i(m-1)}$ on C with the vertex w_i for all $i = 0, 1, \dots, \ell-1$ produces one induced n -cycle containing every w_i .) Thus, $fr(C_m, C_n) \leq n + \lceil n/(m-1) \rceil$.

Case 3. $m-1 \mid n+1$.

Thus $n = \ell(m-1) - 1$. Let $I_{m,n}$ be the graph obtained from $G_{m,n}$ by deleting the edge $w_{\ell-1}v_1$ and adding a new vertex w_ℓ and joining it to v_0, v_2 and $w_{\ell-1}$. Then C_m and C_n can be homogeneously embedded in the graph $I_{m,n}$ of order $n + \ell + 1$. Thus, $fr(C_m, C_n) \leq n + \lceil n/(m-1) \rceil + 1$.

Case 4. $m-1$ does not divide $n-1$ or n or $n+1$.

Thus $n = (\ell-1)(m-1) + r$ for some r satisfying $1 < r < m-2$. Let $J_{m,n}$ be the graph obtained from $G_{m,n}$ by adding a new vertex w_ℓ and joining it to $v_{(\ell-1)(m-1)}$, $v_{(\ell-1)(m-1)-1}$ and $v_{(\ell-1)(m-1)+1}$ (where addition is taken modulo n ; that is, w_ℓ is joined to v_{n-r} , v_{n-r-1} and v_{n-r+1}). Then C_m and C_n can be homogeneously embedded in the graph $J_{m,n}$ of order $n + \ell + 1$. Thus, $fr(C_m, C_n) \leq n + \lceil n/(m-1) \rceil + 1$. $\square$

For large n and fixed m , the upper bounds on $fr(C_m, C_n)$ established in Theorem 2 may be improved considerably. Before presenting another upper bound on $fr(C_3, C_n)$, we note the following integer optimization result.

Lemma 2 *For positive integer b ,*

$$\min_{a \in \mathbb{N}} a + \left\lceil \frac{b}{a} \right\rceil = \left\lceil 2\sqrt{b} \right\rceil,$$

and the minimum is attained when a is the nearest integer to $\sqrt{b}$.

Theorem 3 For all integers $n \geq 8$,

$$fr(C_3, C_n) \leq n + \lceil 2\sqrt{n-3} \rceil - 2.$$

Proof. For $a \geq 3$ an integer, let G_n^a be the graph defined as follows. Start with an induced cycle C on $n + a - 3$ vertices: $v_0, v_1, \dots, v_{n+a-4}, v_0$. Then add $k = \lceil (n + (a - 3))/a \rceil$ new vertices $w_0, w_1, \dots, w_{k-1}$ and for $i = 0, 1, \dots, k - 1$ join w_i to the a vertices $v_{ai}, v_{ai+1}, \dots, v_{a(i+1)-1}$ (where addition is taken modulo $n + a - 3$). Then each vertex of G_n^a clearly belongs to a C_3 . Furthermore, G_n^a homogeneously embeds C_n . To see this, observe that each vertex w_i belongs to an induced C_n namely $w_i, v_{a(i+1)-1}, v_{a(i+1)}, \dots, v_{ai}, w_i$. Hence C_3 and C_n can be homogeneously embedded in the graph G_n^a of order $f(a) = n - 3 + a + k = n - 2 + a + \lceil (n - 3)/a \rceil$.

We minimize this quantity subject to the constraint of n fixed. By Lemma 2, this is obtained when a is the nearest integer to $\sqrt{n-3}$, whence $f(a) = n + \lceil 2\sqrt{n-3} \rceil - 2$ for $n \geq 10$. For $n = 8, 9$ the upper bound is attained for $a = 3$. $\square$

Before presenting the next upper bound, we note the following integer optimization result.

Lemma 3 For positive integers μ and ν ,

$$\min_{a \in \mathbf{N}} a\mu - 1 + \frac{\nu}{a\mu - 1} = 2\sqrt{\nu} + O\left(\frac{\mu^2}{\sqrt{\nu}}\right).$$

Moreover, if $(\sqrt{\nu} + 1)/\mu$ is integral, then the minimum is exactly $2\sqrt{\nu}$.

Proof. Let $f(a) = a\mu - 1 + \nu/(a\mu - 1)$. Real optimization by calculus yields that $f(a) \geq 2\sqrt{\nu}$ attained when $a = \alpha = (\sqrt{\nu} + 1)/\mu$. For an upper bound on integer optimization, we take $a = \lceil \alpha \rceil$. So $a = \alpha + \delta$ for some δ satisfying $0 \leq \delta < 1$. By Taylor's theorem, $f(\alpha + \delta) = 2\sqrt{\nu} + R(\delta)$ where the remainder term satisfies $|R(\delta)| \leq (\mu^2 \delta^2)/(4\sqrt{\nu})$ since $f'(\alpha) = 0$ and $f''(a) = \mu^2 \nu / (2(a\mu - 1)^3)$ is decreasing. The result follows as $\delta < 1$. $\square$

Theorem 4 For integers $n > m \geq 4$,

$$fr(C_m, C_n) \leq n - m + \lceil 2\sqrt{n - m + 1} \rceil + O\left(\frac{m^2}{\sqrt{n}}\right).$$

Moreover, if $(\sqrt{n - m + 1} + 1)/(m - 2)$ is integral, then

$$fr(C_m, C_n) \leq n - m + 2\sqrt{n - m + 1} + 2.$$

Proof. For $a \geq 2$ an integer, let $F_{m,n}^a$ be the graph defined as follows. Start with an induced cycle C on $n + (a - 1)(m - 2) - 2$ vertices: $v_0, v_1, \dots, v_{n+(a-1)(m-2)-3}, v_0$. Let

$$k = \left\lceil \frac{n + (a - 1)(m - 2) - 2}{a(m - 2) - 1} \right\rceil = 1 + \left\lceil \frac{n - m + 1}{a(m - 2) - 1} \right\rceil.$$

Figure 3: The graph $F_{5,20}^3$

Now add k vertices $w_0, w_1, \dots, w_{k-1}$, and for $i = 0, 1, \dots, k-1$ join w_i to the a vertices $v_{i(a(m-2)-1)+j(m-2)}$ for $0 \leq j \leq a-1$, as well as to w_{i-1} and w_{i+1} (where addition is taken modulo $n + (a-1)(m-2) - 2$). The graph $F_{5,20}^3$ is depicted in Figure 3.

Then $F_{m,n}^a$ homogeneously embeds C_m —the induced m -cycles are either of the form w_i , two consecutive neighbours of w_i on C , and the segment of C in between; or of the form w_i, w_{i+1} , the last neighbour of w_i on C , the first neighbour of w_{i+1} on C , and the segment of C in between. Furthermore $F_{m,n}^a$ homogeneously embeds C_n —the induced n -cycles are of the form w_i , its first and last neighbours on C , and the long segment of C joining them. Hence C_m and C_n can be homogeneously embedded in the graph $F_{m,n}^a$ of order $g(a) = n + (a-1)(m-2) - 2 + k$. Thus,

$$g(a) = n - m + 2 + (a(m-2) - 1) + \left\lceil \frac{n - m + 1}{a(m-2) - 1} \right\rceil.$$

We minimize this quantity subject to the constraint of m and n fixed. By Lemma 3, an upper bound on this minimum is obtained when a is the nearest integer to $(\sqrt{n-m+1} - 1)/(m-2)$ whence

$$\min_{a \in \mathbf{N}} g(a) = n - m + 2\sqrt{n-m+1} + O\left(\frac{m^2}{\sqrt{n}}\right).$$

Moreover, if $(\sqrt{n-m+1}+1)/(m-2)$ is integral, then the minimum is $n-m+2\sqrt{n-m+1}+2$. $\square$

4 Lower Bounds

In this section we investigate lower bounds on $fr(C_m, C_n)$. We begin with a lower bound on $fr(C_3, C_n)$.

Theorem 5 *For all integers $n \geq 4$,*

$$fr(C_3, C_n) \geq n + 2\sqrt{n+6} - 5.$$

Proof. Suppose C_3 and C_n can be homogeneously embedded in the graph H . Consider an induced n -cycle C . Since each vertex belongs to a 3-cycle, each vertex on C has a neighbour outside C . For each vertex of C we associate it with one of its exterior neighbours. Define for a vertex w in $T = H - C$ the associativity $a(w)$ of w as the number of associates that it has. The sum of these associativities obeys:

$$\sum_{w \in T} a(w) = n.$$

Let w^* be a vertex in T with the largest associativity and consider an induced n -cycle D that contains w^* . We first note that D contains at most two neighbours of w^* . Thus, at most two associates of w^* are in D . Now look at any other vertex w in T . If $a(w) > 2$, then the cycle D must miss either w or at least one of its associates (by the same argument).

Let t denote the number of vertices in T and y the number of vertices of T with associativity exceeding 2. If $y > 0$, this means that the number of vertices not in the cycle D —which is equal to t since C and D have the same length—is at least

$$t \geq (y - 1) + (a(w^*) - 2). \quad (1)$$

(If $y = 0$ then $t \geq n/2$ so that $p(H) \geq 3n/2$ which exceeds the claimed lower bound.) Since $t - y$ vertices of T have associativity at most 2, and since the sum of the associativities is n , there are at least $n - 2(t - y)$ vertices of C that are associated with the y vertices of T whose associativity is at least 3. By our choice of w^* , $a(w^*) \geq (n - 2(t - y))/y$. Substituting into (1), we obtain that

$$t \geq y - 1 + \frac{n - 2t}{y}.$$

Multiplying out and solving for t yields:

$$t \geq \frac{y^2 - y + n}{y + 2}.$$

Real optimization by calculus yields $y^* = \sqrt{n + 6} - 2$. Thus

$$p(H) \geq n + 2\sqrt{n + 6} - 5. \quad \square$$

Next we present a lower bound on $fr(C_m, C_n)$ for $m \geq 4$ but small.

Theorem 6 *For integers $n > m \geq 4$ with $m \leq \sqrt{n}/4$,*

$$fr(C_m, C_n) \geq n + 2\sqrt{n + 2m} - m - 2.$$

Proof. Let H be a graph of order $fr(C_m, C_n) = n + t$ in which C_m and C_n can be homogeneously embedded. We show that $t \geq 2\sqrt{n + 2m} - m - 2$. Let C be any induced n -cycle. We refer to those vertices of H not in C as exterior vertices, and we let T denote the set of all exterior vertices. Thus $|T| = t$.

We partition ownership of the graph amongst the exterior vertices as follows. An exterior vertex owns itself. In order to assign ownership of the vertices of C to the exterior vertices, we colour the vertices of C red or green depending on whether they have an exterior neighbour or not. Thus each green vertex has degree 2 in H , while each red vertex has degree at least 3. We will also sometimes split vertices in two so that two exterior vertices each own half of the vertex.

Let R denote the set of red vertices of C . We define each component of $C - R$ to be a *tract*. That is, a tract is a maximal connected subgraph consisting of green vertices. We

split the ownership of the vertices of each tract in the middle and call each half a *demitract*. For example, if the tract is $x_1x_2x_3$ the one demittract consists of x_1 and half of x_2 , and the other demittract consists of x_3 and half of x_2 .

Each red vertex that is adjacent with a green vertex is called the *red end* of the demittract containing that green vertex. If two demitracts have the same red end, then they and the red end are assigned to an exterior neighbour of the red end. Such an exterior vertex we call *rich*.

Each remaining demittract is then assigned to some neighbour of the red end. What remains to assign are some of the red vertices.

We now define an exterior vertex as *poor* if it is not rich and it owns either one or two demitracts. If a vertex w is poor, then for each demittract that w owns it makes a claim on two red vertices: the red end of the demittract that it is adjacent to and the vertex adjacent to that red end. So it is possible for a red vertex to be claimed by up to two poor vertices. If a red vertex is claimed by one poor vertex w , then w becomes its owner. If a red vertex is claimed by two poor vertices, then the two poor vertices each own half the red vertex.

Each remaining red vertex that does not yet have an owner is assigned to some exterior neighbour. The total ownership is $n + t$, the number of vertices of H . We now prove the following claim.

Claim *Let D be any induced cycle in H and let $v \in T$.*

- (i) *If D contains all of v 's possessions, then v owns at most three vertices.*
- (ii) *If v is in D , then D contains at most m vertices that v owns.*

Proof. (i) If v is rich, then v owns a red end and part of three neighbours of the red end, namely v itself and the two green vertices of the demitracts adjacent to the red end. These cannot simultaneously be in an induced cycle.

Suppose that D contains all of v 's possessions. Then v must belong to D . So D includes only two neighbours of v . In particular, D can include at most two demitracts owned by v .

If v is poor, then consider the red end of a demittract owned by v (so that the red end is adjacent to v). Now v owns part of the red end and three neighbours of the red end, namely v itself, the red vertex adjacent to the red end and the green vertex of the demittract adjacent to the red end. These four vertices cannot simultaneously be in an induced cycle. Hence v does not own any demittract.

Thus the only vertices v owns are red neighbours. These must all be in D . Hence the total ownership of v is at most 3.

(ii) Note that a tract has length at most $m - 3$, since each of its vertices is in an m -cycle. Thus v possesses at most $\mu = (m - 3)/2$ vertices from each demittract that it owns. Suppose now that v belongs to D . Then at most two demitracts owned by v belong to D . Thus v 's total ownership in D is at most $2\mu + 3 = m$. $\square$

For each vertex v in T , let $own(v)$ be the number (or rather the weight) of vertices owned by v . Let Z denote the set of all exterior vertices which own at least $m + 1$ vertices, and let $|Z| = z$. Furthermore, let Y denote the set of all exterior vertices who own more than 3 but less than $m + 1$ vertices, and let $|Y| = y$. Hence each $y \in Y$ owns at least three and a half vertices and at most m and a half vertices. If $z = 0$ then $t > n/(m + 1/2)$, and we are done. So assume otherwise.

We now consider any $w \in Z$ and any induced cycle D containing w . We count the number of vertices that are not in D . By Claim (ii), at least $own(w) - m$ vertices owned by w are not in D , and at least one vertex owned by each remaining vertex of $Z - \{w\}$ is not in D . For each $y \in Y$, D cannot contain all of y 's possessions by Claim (i). Hence at least half a vertex owned by each $y \in Y$ is not in D . Thus, the number of vertices that are not in D is at least

$$y/2 + (z - 1) + own(w) - m.$$

The total ownership by vertices in $T - Z$ is at most $3(t - y - z) + (m + 1/2)y$. It follows that if we take w^* to be a vertex of Z with the largest ownership, then

$$own(w^*) \geq \frac{(n + t) - (3(t - y - z) + (m + 1/2)y)}{z}.$$

But the number of vertices not in an induced n -cycle is equal to t . And w^* is in an induced n -cycle. Hence

$$t \geq y/2 + z - 1 - m + \frac{(n + t) - (3(t - y - z) + (m + 1/2)y)}{z}.$$

Solving for t :

$$t \geq \frac{z^2 + n + 2z - mz + y(z + 5 - 2m)/2}{z + 2}$$

But also, $t \geq y + z$. Hence, to obtain a lower bound for t we must compute the following:

$$\min_{y,z} \max\{z + y, A(z) + B(z)y\}$$

where $A(z) = (z^2 + n + 2z - mz)/(z + 2)$ and $B(z) = (z + 5 - 2m)/(2(z + 2))$. Consider a fixed value of z . If $B(z) > 0$, then both functions inside the maximum are increasing in y . Hence the minimum is achieved at $y^* = 0$. On the other hand, if $B(z) < 0$, then $z + y$ is increasing in y but $A + By$ is decreasing in y . Hence the minimum value of the maximum occurs at the intersection; i.e., where $y = (A - z)/(1 - B)$. Hence

$$t \geq \min \left\{ \min_z A(z), \min_z \frac{A(z) - zB(z)}{1 - B(z)} \right\}.$$

It is straightforward (but tedious) to see that $A''(z) > 0$ always, and that $A(z)$ is minimized when $z^* = \sqrt{n + 2m} - 2$. In this case it follows that

$$A(z) \geq \alpha = 2\sqrt{n + 2m} - m - 2.$$

The second function $(A - zB)/(1 - B)$ above simplifies to

$$\frac{z^2 - z + 2n}{z + 2m - 1}.$$

The second derivative of this function is always positive, and the function's minimum is attained at $z^* = \sqrt{2}\sqrt{n + 2m^2 - m} - 2m + 1$. Hence

$$\frac{A(z) - zB(z)}{1 - B(z)} \geq \beta = \sqrt{8}\sqrt{n + 2m^2 - m} - 4m + 1.$$

Now, we need to compare α with β . It follows that

$$\begin{aligned} \beta - \alpha &> \sqrt{8}\sqrt{n + 2m^2 - m} - 2\sqrt{n + 2m} - 3m \\ &> (\sqrt{8} - 2)\sqrt{n + 2m} - 3m \\ &> (\sqrt{8} - 2)\sqrt{n} - 3m. \end{aligned}$$

The latter quantity is equal to 0 if $m = (\sqrt{8} - 2)\sqrt{n}/3 \approx 0.276\sqrt{n}$. It follows that $\alpha < \beta$ for at least the range $m \leq \sqrt{n}/4$. Hence, for this range of m ,

$$t \geq \alpha = 2\sqrt{n + 2m} - m - 2. \quad \square$$

We present next a lower bound on $fr(C_m, C_n)$ for m large.

Theorem 7 *For all integers $n > m \geq \sqrt{6n}$,*

$$fr(C_m, C_n) \geq n + n/m.$$

Proof. Let H be a graph of order $fr(C_m, C_n) = n + t$ in which C_m and C_n can be homogeneously embedded. If $m < 6t$, then $t > m/6 \geq n/m$ since $m \geq \sqrt{6n}$. Hence in what follows we may assume that $m \geq 6t$ for otherwise the result is immediate.

Let C be any induced n -cycle. We will refer to those vertices of H not in C as exterior vertices, and we let T denote the set of all exterior vertices. Let S denote the set of vertices on C with exterior neighbours. We now introduce the notion of *lanes* and *sentries* in C .

We define a component of $C - S$ to be a *lane* if it has more than t vertices. Let x denote the number of lanes, and let X denote the set of vertices of C that belong to a lane. Each vertex in a lane is in an induced m -cycle. So it follows that a lane has at most $m - 3$ vertices. Hence

$$|X| \leq x(m - 3). \quad (2)$$

Next we define a vertex of S as a *sentry* if it is adjacent to a vertex of a lane. For each sentry s we choose one external neighbour e_s . Any n -cycle contains at least one vertex from each lane. Thus every n -cycle contains every sentry. In particular, an induced n -cycle that contains e_s must use the edge se_s . We call that edge *essential*.

It follows that an exterior vertex is incident with at most two essential edges and in fact is adjacent to at most two sentries. Thus, there are at most $2t$ sentries in total. Each lane

has two sentries. A sentry s cannot be adjacent to two lanes (since an n -cycle containing e_s would be forced to use s and both lanes). Thus there are exactly $2x$ sentries. In particular,

$$x \leq t.$$

We now count the number of vertices of C that do not belong to a lane. Let A denote the set of vertices of C that are sentries or surrounded by sentries (i.e., both neighbours on C are sentries). Then

$$|A| \leq 3x. \quad (3)$$

Let B denote the remaining vertices of C that do not belong to A nor to a lane. We now associate each vertex of B with an exterior vertex. For each vertex in B that has an exterior neighbour (that is, $B \cap S$), we associate it with one such neighbour. Consider a maximal segment of $B - S$. We call such a segment a tract. As in the above proof, it is helpful to split a vertex in two. Thus one can split the tract into two *demitracts*. Each demittract has a unique vertex of S that is its *end*. Associate each demittract with the vertex with which its end is associated. This process partitions B .

For an exterior vertex w , let F_w denote the set of vertices and half-vertices of B associated with w . By $|F_w|$ we mean the weight of the vertices associated with w , each vertex associated with w counting one and each half-vertex counting half. We now bound $|F_w|$. To do this, we first partition the set T of exterior vertices as follows.

We say that an exterior vertex is *predictable* if it is incident with two essential edges, otherwise it is *unpredictable*. Let T_p (T_{up}) denote the set of all predictable (respectively, unpredictable) vertices, so $T = T_p \cup T_{up}$. Since each sentry is incident with exactly one essential edge, there are $2x$ essential edges. However, the number of essential edges is at most $2|T_p| + |T_{up}| = |T| + |T_p| = t + |T_p|$. Thus, $2x - t \leq |T_p|$. Hence,

$$|T_{up}| = |T| - |T_p| \leq 2(t - x).$$

Suppose now that w is predictable. We show then that $|F_w| \leq 2(t - x)$. Consider an induced n -cycle D that contains w . Since w is predictable, D does not contain any vertex of B that is a neighbour of w . Furthermore, it cannot contain a demittract of $B - S$ associated with w . Hence D contains none of F_w .

Let a denote the number of vertices in C but not D ; so D contains a exterior vertices. Let T'_p (T'_{up}) denote the set of all predictable (respectively, unpredictable) vertices in D , so $|T'_p| + |T'_{up}| = a$. Let J denote the sentries whose essential edges are in D . Each essential edge in D joins a vertex of J with one of the exterior vertices in D . Thus there are $|J|$ essential edges in D . On the other hand, the number of essential edges in D is $2|T'_p| + |T'_{up}| = 2a - |T'_{up}| \geq 2a - |T_{up}| \geq 2a - 2(t - x)$. Thus,

$$|J| \geq 2a - 2(t - x).$$

If an element of $C - D$ is in B , then it is not surrounded by J . Hence at most one of its neighbours is in J . On the other hand, each sentry in J has a neighbour on C which is not

in D . If we let f denote the number of vertices of $C - D$ that are not surrounded by J , then $f + 2(a - f) = |J|$. Thus,

$$f + 2(a - f) = |J| \geq 2a - 2(t - x).$$

That is, $f \leq 2(t - x)$. In particular, since $|F_w| \leq f$,

$$|F_w| \leq 2(t - x).$$

On the other hand, suppose that w is unpredictable. Consider an induced n -cycle D that contains w . The cycle D can visit at most two neighbours of w . Therefore it can visit at most two demitracts associated with w . The rest of F_w must be part of the t vertices not in D . It follows that

$$|F_w| \leq 2t + 2.$$

Hence

$$|B| = \sum_{w \in T_p} |F_w| + \sum_{w \in T_{up}} |F_w| \leq |T_p| \cdot 2(t - x) + |T_{up}| \cdot (2t + 2).$$

Since $|T_p| \leq t$ and $|T_{up}| \leq 2(t - x)$, it follows that

$$|B| \leq 6t(t - x). \quad (4)$$

Using equations (2), (3) and (4), we now have a bound on the number $|A| + |B| + |X|$ of vertices of C , namely,

$$n \leq x(m - 6t) + 6t^2.$$

Since $m \geq 6t$, the expression $x(m - 6t) + 6t^2$ is maximized at x maximum; viz. when $x = t$. In this case it follows that

$$t \geq n/m. \quad \square$$

5 Conclusions

What happens for the range of m between $\sqrt{n}/4$ and $\sqrt{6n}$ is unclear. The upper bounds of Section 3 apply, and to obtain lower bounds one can use techniques similar to those of Section 4. However, neither the lower nor the upper bound seems particularly sharp for this range of m , and the values of $fr(C_m, C_n)$ seem to depend considerably on whether n is (about) a multiple of m or not.

The following table summarizes the results of this paper.

Range	$fr(C_m, C_n)$
$3 \leq m \leq n^{1/4}$	$n - m + 2\sqrt{n} \pm O(1)$
$n^{1/4} \leq m \leq \sqrt{n}/4$	between $n - m + 2\sqrt{n}$ and $n - m + 2\sqrt{n} + O(m^2/\sqrt{n})$
$\sqrt{n}/4 \leq m \leq \sqrt{6n}$	between $n + \Omega(\sqrt{n})$ and $n + O(\sqrt{n})$
$m \geq \sqrt{6n}$	$n + \frac{n}{m} + O(1)$

As a consequence of Theorems 3, 4, 5 and 6 we obtain:

Corollary 1 *For m fixed and n large, $fr(C_m, C_n) = n - m + 2\sqrt{n} \pm O(1)$.*

As a consequence of Lemma 1 and Theorems 1 and 2 we obtain:

Corollary 2 *For $7 \leq m + 2 \leq n \leq 2m - 5$, $fr(C_m, C_n) = n + 3$.*

As a consequence of Theorems 2 and 7 we obtain:

Corollary 3 *If $(m - 1)|n$ and $m \geq 6\sqrt{n}$, then $fr(C_m, C_n) = n + \frac{n}{m-1}$.*

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